

ABBY - document classification assignment

Description:

Neural networks are usually trained on large-scale labeled datasets. As labeling data is a slow and costly process, it is crucial to wisely select the documents to annotate.

The task is to implement a method for flagging which documents to annotate for a document classification task.

It should be performed on the [DocILE dataset \(GitHub\)](#). It is assumed that only the documents in the provided training set are annotated, and the task is to select additional documents from the provided unlabeled pool for which the annotation can be used for training, such that we aim to achieve the best possible accuracy on the validation set. Validate your solution for the unsupervised/semi-supervised document flagging on the validation set by comparing with relevant baselines and write a documentation about the experiments.

Detailed tasks:

- Register and download the annotated-trainval dataset from the link provided above. The dataset is a compressed zip file containing the training and validation documents and labels, for this experiment, use the validation (val.json > new_train.json) set as training set and the training (train.json > unlabeled.json) set as the pool of unlabeled documents.
- Implement a neural network in PyTorch or TensorFlow for the document classification task using a VLM / LLM with fewer than 3 billion parameters or use a publicly available implementation.
- Train the first model on the whole dataset as a baseline: use this only as a performance ceiling (do not use it to select or flag documents for annotation).
- Train the second model on the training set only: this is the baseline we're trying to improve.
- Select 1500 documents from the unlabeled pool (without looking at their labels) for annotation.
- Train the third model on the training set plus the extra documents from the unlabeled pool, after retrieving their annotations.
- Compare results obtained from various selection methods. Write a documentation about the selection methods used in the experiments and discuss the results and share that with us via hiring-ailabs@abby.com
- In case you have questions regarding the assignment, use the same email for reaching out: hiring-ailabs@abby.com